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Enhanced anaerobic digestion of rice straw by using microbial pretreatment augmented with micro-nano bubble water

Suyun Xu [a,*], Yakun Lu [a], Xuezhi Liu [a], Ruixiang Zou [a], Yuchen Zhang [a], Mukesh Kumar Awasthi [b], Hewen Wei [c], Hongbo Liu [a,*]

[a] School of Environment and Architecture, University of Shanghai for Science and Technology, Shanghai, 200093, China
[b] College of Natural Resources and Environment, Northwest A&F University, Yangling, Shanxi Province, 712100, China
[c] Shanghai Microcre Optics-Mech Tech Co., Ltd, Shanghai, 201411, China

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ABSTRACT

Micro-nano bubbles (MNBs) can enhance microbial processes, but their role in boosting defined microbial consortia for biomass pretreatment is unknown. This study demonstrates a novel pretreatment strategy that synergistically combines compound microbial consortia with MNBs to dramatically improve rice straw decomposition and anaerobic digestion. The compound consortia were comprised of *Bacillus velezensis* HAB-2 (S1), *Phanerodontia chrysosporium (S2)*, and *Phanerodontia chrysosporium* SILI (S3) at varied ratios. Based on straw degradation rate, hydrolysate sCOD, and methane production, the proportion of microbial consortia differentially impacted the degradation of cellulose, hemicellulose, and lignin. Specifically, the K-1 (S1:S2:S3 = 1:1:1) and K-2 (S1:S2:S3 = 2:1:1) consortia showed the best synergistic effects. K-2 consortium with Air-MNBs achieved the highest degradation rates of total solids (TS, 43.5 %) and volatile solids (VS, 28.0 %), which increased 58.2 % and 86.7 %, respectively over controls without MNBs injection. Correspondingly, the K-2 consortium pretreated with Air-MNBs achieved a maximum methane yield of 47.5 mL/g VS, significantly higher than that of the untreated control group. Air-MNBs were more effective than N_2-MNBs in bacterial pretreatment, whereas their enhancement in fungal pretreatment was not significant. This work establishes a new approach for enhancing lignocellulosic biomass pretreatment by harnessing the synergy between designed microbial consortia and MNBs technology.

1. Introduction

Biomass energy is a crucial renewable resource, and agricultural waste such as rice straw serves as an important biomass source (Wang et al., 2022). Globally, approximately one billion tons of straw are produced annually (Harun et al., 2022). Improper disposal not only wastes energy but also occupies space and causes environmental damage. A promising solution is converting straw into biogas through anaerobic digestion (AD), which can help to alleviate energy shortages. As a lignocellulosic material, straw contains 32–47 % cellulose, 19–27 % hemicellulose, and 5–24 % lignin (Harun et al., 2022). Its recalcitrant structure, characterized by lignin encapsulation and silicified surfaces, hinders microbial degradation of cellulose and makes hydrolysis the rate-limiting step in AD (Sawatdeenarunat et al., 2015). Some studies have tried the utilization of the co-culture of microbial agents to enhance the degradation of rice straw, with the degradation rates of lignin and cellulose being 26.38 % and 33.29 %, respectively (Chen et al., 2019). However, achieving a high and cost-effective degradation rate of lignocellulose in straw to substantially boost biogas production remains a significant challenge.

Micro-nano bubbles (MNBs), defined as bubbles with diameters below 100 μm, exhibit unique properties such as prolonged stability, high specific surface area, and efficient mass transfer (Xiao et al., 2025). Aeration with MNBs increases dissolved oxygen levels, supporting microbial growth and organic pollutant degradation (Lyu et al., 2019). The high zeta potential of MNBs enhances particle dispersion and substrate accessibility, thereby improving methane production (Chuenchart et al., 2021; Hou et al., 2024). Furthermore, MNBs influence enzyme activity through surface charge interactions and generation of hydroxyl radicals (•OH) during collapse, which may alter enzyme conformation or redox state (Yamaguchi et al., 2021; Zhang et al., 2017). The sustained oxygen release also promotes aerobic microbial metabolism and associated

* Corresponding authors.
E-mail addresses: xusy@usst.edu.cn (S. Xu), Liuhb@usst.edu.cn (H. Liu).

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enzymatic pathways, underscoring the dual role of MNBs as physical and biochemical modulators in enzyme-mediated processes (Silva et al., 2024).

Biological pretreatment using microbial consortia represents a promising avenue for enhancing lignocellulose deconstruction. The synergy between fungal and bacterial members in composite microbial agents can lead to more effective degradation of straw components than single strains. This is evidenced by recent studies; for instance, a defined microbial consortium was reported to increase cellulolytic enzyme activities significantly and achieve a 45.9 % enhancement in biogas production from rice straw (Sahil et al., 2023). Furthermore, the combination of biological and physicochemical pretreatments has been shown to be a highly effective strategy. For instance, the integration of ultrasonic pretreatment with bioaugmentation achieved the highest methane production and significantly enhanced the crude cellulose removal rate during anaerobic digestion, demonstrating a clear synergistic effect (Zhang et al., 2023). Despite these advances, biological pretreatment efficiency is often hampered by intrinsic limitations including slow mass transfer and suboptimal microbial synergy within the consortia.

Concurrently, MNBs technology has demonstrated considerable potential in enhancing AD processes. Its application has been shown to substantially improve methane yield from various organic wastes. For example, adding Air-/H_2-/CO_2-nanobubble water to swine manure digestion increased cumulative methane production by 19–39 %, attributed to accelerated hydrolysis of proteins/carbohydrates and enhanced enzyme activity (Fan et al., 2020). In the context of waste activated sludge, supplementation with N_2-nanobubble water boosted methane yield by 29 %, which was mechanistically linked to a 14–17 % increase in the activities of key extracellular hydrolases (Yang et al., 2019). Critically, while MNBs have been used within the digestion phase or as a soaking agent (Wang et al., 2023), their potential to specifically augment a dedicated microbial pretreatment stage has been completely overlooked. This represents a key strategic gap. We posit that using MNBs to create an optimized microenvironment during pretreatment can powerfully amplify the synergy and enzymatic activity of inoculated consortia.

It is hypothesized that the unique properties of MNBs (e.g., high mass transfer, reactive oxygen species generation, and zeta potential modulation) can create a superior micro-environment that amplifies the activity and synergy of pre-inoculated, specialized consortia. To bridge this gap, this study systematically investigates the synergistic effects of combining specifically formulated microbial consortia (*Bacillus velezensis* and *Phanerodontia chrysosporium* at different ratios) with different types of MNBs water (Air vs. N_2) for pretreating rice straw. The objectives are to: (1) evaluate the enhancement in straw decomposition efficiency; (2) elucidate the underlying mechanisms through enzyme activity profiling; and (3) determine the ultimate impact on biomethane production. This work is distinguished by three key innovations: (1) We are the first to employ MNBs as a dedicated enhancer for a standalone microbial pretreatment stage, rather than as a direct additive to anaerobic digestion. (2) We systematically investigate the interaction between specifically formulated bacterial-fungal consortia and different types of MNBs (Air vs. N_2). (3) We move beyond performance metrics to elucidate the underlying mechanisms through extracellular enzyme activity profiling.

2. Materials and methods

2.1. Screening of strains

The substrate for AD was rice straw, which was air-dried and ground to an average particle size of 1–2 cm before use. The anaerobic inoculum was digested sludge obtained from a full-scale mesophilic biogas plant (Ouhai Energy Co., Ltd., Chongming District, Shanghai, China). To minimize endogenous gas production, the inoculum was pre-incubated at 37 °C in the laboratory until gas production ceased, and then stored at 4 °C prior to use. The fundamental physicochemical properties of the rice straw and inoculum were characterized, and the detailed data are provided in Table S1 (Supplementary Information).

The bacterial strains were isolated from compost samples using a cellulose-Congo red medium to screen for cellulose-degrading ability (Mathews et al., 2014). The screened strains were further isolated, purified, and expanded in culture. Subsequently, they were identified by 16S rDNA and ITS sequencing. Strain S1 was identified as belonging to the genus *Bacillus*, showing 100 % similarity to *Bacillus velezensis strain HAB-2*. Strains S2 and S3 were identified as belonging to the genus *Phanerodontia*, exhibiting 99.84 % similarity to *Phanerodontia chrysosporium* and 99.81 % similarity to *Phanerodontia chrysosporium SILI*, respectively.

Three strains were screened by the Congo red heating color method. The screened strains all exhibited relatively high cellulase activities in the culture medium, which were 5.49 U/mL, 5.26 U/mL, and 4.73 U/mL respectively. Through experiments on aniline blue medium and guaiacol medium, it was qualitatively verified that both strains S2 and S3 could produce laccase and lignin peroxidase, indicating their ability to degrade lignin. Preparation of spore suspension: The isolated strain was activated and inoculated onto PDA plates, which were then incubated at 28 °C for 5–7 days. Once the colonies covered the entire medium surface, spores were rinsed from the plate surface into an Erlenmeyer flask using sterile saline solution. The suspension was shaken at 120 r/min for 3 h, followed by spore counting using a hemocytometer (Hennechart-Collette et al., 2014), and the concentration of the bacterial suspension of each strain was adjusted to 10^8 CFU/mL with sterile distilled water as the initial inoculation dose for the pretreatment. Throughout the 18-day pretreatment period, this initial dose ensured a substantial and active microbial population to initiate straw decomposition, with the subsequent dynamics driven by microbial growth and metabolic activity on the straw substrate.

2.2. Experimental design

This study was conducted in three sequential phases to systematically evaluate the pretreatment of rice straw. The basic pretreatment procedure for all batches was as follows: 5 g of rice straw was added into 110 mL of deionized water, and then sterilized at 121 °C for 15 min in an autoclave (Model 2018-B3285, 30 L capacity, Shanghai Boxun Medical Biological Instrument Corp, China). Subsequently, specific microorganisms and/or MNBs modifiers were added as described below, followed by 18 days of pretreatment prior to anaerobic digestion.

Phase I: Screening of Microbial Consortium Ratios.

This phase aimed to identify the most effective ratio of the three strains (S1, S2, S3) without MNBs augmentation. H-1, H-2, H-3 groups: received 60 mL of mixed microbial suspension (10^8CFU/mL) at ratios of S1:S2:S3 = (1:1:1; Balanced), (2:1:1; Bacterium-Enhanced), and (1:2.5:2.5; Fungus-Dominated), respectively. Control (CK): received 60 mL of sterile deionized water instead of microbial suspension.

Phase II: Augmentation of the Optimal Consortium with Air-MNBs.

Based on the results of Phase I, this phase assessed the synergy between the optimal consortium and Air-MNBs. K-1, K-2, K-3 groups: corresponded to the same microbial ratios as H-1, H-2, and H-3, but pretreated with a mixture of 60 mL bacterial suspension and 100 mL Air-MNBs water. Control (CK-1): received 100 mL Air-MNBs water and 60 mL sterile water without microbes.

Phase III: Comparison of MNBs Types on Functional Groups.

This phase dissected the effects of different MNBs types (Air vs. N_2) on the key functional units. A-1: S1 augmented with Air-MNBs; N-1: S1 augmented with N_2-MNBs; A-2: fungal consortium (S2:S3 = 1:1) augmented with Air-MNBs; N-2: fungal consortium (S2:S3 = 1:1) augmented with N_2-MNBs. The pretreatment medium for all Phase III groups was 60 mL of the respective microbial suspension mixed with 100 mL of the specified MNBs water.

MNBs Water Generation: MNBs water for all phases was prepared using a generator (ZJC-NM, Shanghai Zhongjing Environmental Protection Technology Co., Ltd.). Deionized water was aerated with compressed air or N_2 at 1.0 L/min for 30 min, producing bubbles primarily in the sub-micron range (100–800 nm) at a concentration of $\sim 10^7$–10^8 particles/mL.

Anaerobic Digestion: Following the 18-day pretreatment, 100 mL of anaerobic inoculum was added to each reactor to initiate AD, during which gas production and digestate characteristics were monitored.

2.3. Analytical methods

The samples were heated to a constant weight at 105 °C and 600 °C, respectively, to measure the total solids (TS) and volatile solids (VS). The cellulose content was determined by the sulfuric acid-potassium dichromate oxidation method. The hemicellulose content was measured by the dinitrosalicylic acid (DNS) colorimetric method after hydrolysis with 2 mol/L HCl. The lignin content was determined by the sulfuric acid washing method (Chen et al., 2019).

To test various indicators of the straw hydrolysate, the samples were centrifuged (×14,000 rpm) and then filtered through a 0.45 μm filter membrane. The oxidation-reduction potential (ORP) was measured using an ORP meter (DZS-706-A, Leici, Shanghai). The soluble chemical oxygen demand (sCOD) was determined using the Hach rapid digestion method (DRB200 Hach digester, HACH Company, USA). The polysaccharide content was determined by the phenol-sulfuric acid method and the protein content was measured by the Coomassie Brilliant Blue G-250 spectrophotometric method. The concentrations of volatile fatty acids (VFAs) were quantified by gas chromatography (Agilent 7890) using a flame ionization detector (FID) and an HP-5 capillary column (30 m × 0.32 mm × 0.25 μm). The operational parameters were as follows: helium carrier gas at 1.0 mL/min, an injector temperature of 250 °C, and a detector temperature of 240 °C. A standard solution containing a mixture of VFAs (acetic acid, propionic acid, iso-butyric acid, butyric acid, iso-valeric acid, and valeric acid) was used.

The activities of extracellular enzymes in the fermentation broth were semi-quantitatively profiled using API ZYM test strips (bioMérieux, France). Following the manufacturer's guidelines with modifications, the supernatant of the sample was obtained by centrifugation at 12,000 rpm for 20 min and subsequent filtration through a 0.45 μm membrane. Then, 65 μL of the prepared supernatant was inoculated into each cupule of the API ZYM strip. The strips were incubated at 37 °C for 4 h in a humidified chamber to prevent evaporation. After incubation, ZYM A (a chromogenic reagent) and ZYM B (a pH-stabilizing reagent) were added sequentially to each cupule. The strips were then allowed to stand for 5 min under strong light to allow for full color development. The resulting color in each cupule was immediately compared to the API ZYM reference chart. Enzyme activity was semi-quantitatively scored on a scale from 0 to 5, where 0 indicates no activity (colorless), 1 represents 5 nmol (very low), 2 represents 10 nmol (low), 3 represents 20 nmol (medium), 4 represents 30 nmol (high), and 5 represents ≥40 nmol (very high) of hydrolyzed substrate.

2.4. Statistical analysis

All experimental data were obtained from triplicate trials and expressed as mean values with standard deviations (mean ± SD). Statistical analyses were performed using SPSS (version 27.0, IBM, USA). For comparisons among multiple treatment groups (for example, microbial consortia ratios H-1 to H-3, MNBs-enhanced groups K-1 to K-3, and MNBs-type groups A-1/N-1/A-2/N-2), one-way analysis of variance (ANOVA) with Tukey's honestly significant difference (HSD) post-hoc test was applied. Significant differences were determined at $p < 0.05$. Similarly, Tukey's HSD post hoc test was employed to analyze differences in straw degradation parameters (including TS, VS, cellulose, hemicellulose, and lignin contents) and methane production. Spearman's rank correlation analysis was performed to examine relationships between key operational parameters (polysaccharides, proteins, and sCOD concentrations) and methane production across different time points. Principal component analysis (PCA) was performed to visualize the multivariate effects of different treatments (MNBs types and microbial consortia ratios) on overall system performance. The statistical significance thresholds were set as: $p < 0.05$ (significant), $p < 0.01$ (highly significant), $p < 0.001$ (extremely significant). Graphical representations were generated using Origin (version 2024, OriginLab, USA).

3. Results and discussion

3.1. Straw hydrolysis in biological pretreatment with single strains VS. compound microbial agent

The hydrolysis performance of straw pretreated with single or mixed microbial consortia was evaluated based on the reduction rates of TS and VS. As illustrated in Fig. 1a, all single-strain treatments significantly enhanced straw degradation compared to the control group, with strain S2 exhibiting the highest degradation rate of 11.0 %, slightly superior to strain S3 (9.0 %) and markedly higher than strain S1 (6.0 %). This difference can be attributed to the more efficient, substrate-specific synergy of fungal-derived cellulases and hemicellulases, where fungi maintain lignocellulose degradation capability through hydrolytic enzyme-dominated systems even under hypoxic conditions, while bacterial degradation typically depends on aerobic environments (Rollig et al., 2025). These findings align with previous studies demonstrating fungi's superior lignocellulose degradation efficiency over bacteria in agricultural waste treatment systems. Combined with experimental data and operational conditions, the results clearly indicate that strains S2 and S3 possess stronger lignocellulose matrix degradation capacity than S1. Based on these findings, experimental designs incorporating varying ratios of bacterial-fungal co-cultures could be implemented to investigate potential synergistic enhancement of straw degradation efficiency.

To evaluate the pretreatment effects of bacterial-fungal consortia with different ratios on straw, the degradation rates of TS and VS by mixed strains are presented in Fig. 1b. Comparative analysis with single-strain pretreatment results (Fig. 1a) reveals that the mixed consortia significantly enhanced straw hydrolysis efficiency. The degradation of both TS and VS was monitored, providing complementary insights. The TS reduction reflects the overall mass loss including inorganic components, while VS reduction specifically indicates the biodegradation of organic matter. Notably, the maximum TS degradation rates reached 27.5 %, 22.5 %, and 19.0 % for H-2, H-3, and H-1, respectively, with H-2 being significantly higher than both H-3 and H-1 ($p < 0.05$). The VS degradation is not prominent enough (Fig. 1b), confirming that the mass loss was primarily due to the consumption of organic components. This trend reveals that overall degradation efficiency depends on a balanced synergy between the fungal and bacterial components, rather than merely on the total fungal proportion.

The underlying functional specialization is elucidated in Fig. 1c, which shows the component-specific degradation rates. Due to the inherent laccase production characteristics of *Phanerodontia* sp. (Luo et al., 2024; Pan et al., 2019), the highest lignin degradation rate was found in the fungal-dominated consortium H-3 (35.5 %), which was significantly greater than all other groups ($p < 0.01$). A more nuanced finding is that the H-2 consortium achieved a significantly higher lignin degradation rate (25.1 %) than H-1 (22.8 %) ($p < 0.05$), despite a lower nominal fungal inoculum ratio (1/2 in H-2 vs. 2/3 in H-1). This apparent paradox underscores that microbial synergy, not just inoculum size, governs function. In the H-2 ratio, the larger initial population of *Bacillus velezensis* (S1) may enhance the ligninolytic activity of the fungi by rapidly consuming and removing soluble sugars and intermediate metabolites released during the initial breakdown of straw. This bacterial "scavenging" could prevent the catabolite repression of fungal

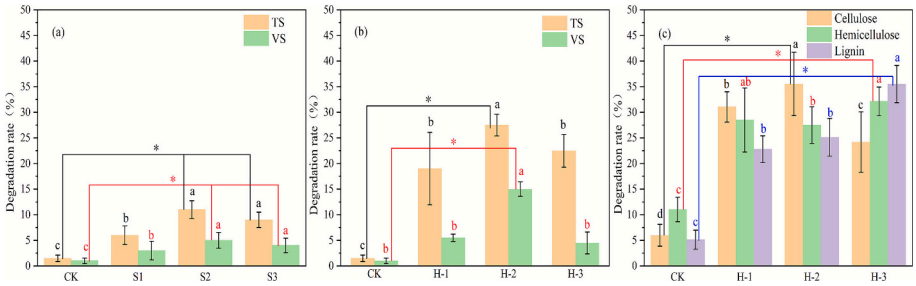


Fig. 1. Straw decomposition in biological pretreatment with (a) single strain and (b, c) compound microbial agent at different ratios: H-1 (S1:S2:S3 = 1:1:1), H-2 (S1:S2:S3 = 2:1:1), and H-3 (S1:S2:S3 = 1:2.5:2.5). The control group (CK) was without microbial inoculation. Data represent mean ± standard deviation ($n = 3$). Bars sharing the same lowercase letter within a panel are not significantly different according to Tukey's honestly significant difference (HSD) post-hoc test ($p < 0.05$).

lignocellulolytic enzymes and maintain a microenvironment that is more favorable for fungal activity, thereby increasing the overall efficiency of lignin decomposition. Conversely, the cellulose degradation efficiency was highest in H-2, indicating that this bacterium-enhanced consortium had a stronger ability to decompose cellulose, which can be attributed to the cellulolytic enzymes of Bacillus sp. (Dong et al., 2024).

These findings demonstrate that the H-2 consortium achieved the best overall performance by fostering a synergistic partnership where bacterial activity potentiated fungal lignin degradation while concurrently executing robust cellulose hydrolysis. The experiment confirms that formulating microbial consortia with complementary functional ratios can significantly promote the degradation of key straw components (Wu et al., 2022).

3.2. Enhanced hydrolysis of straw in biological pretreatment augmented with micro-nano bubbles

The synergistic effects of compound microbial agents augmented with MNBs (Air and N_2, respectively) were explored. As shown in Fig. 2a, with the augmentation of Air-MNBs, the maximum reduction of TS (43.5 %) and VS (28 %) was found in the K-2 group, which was significantly higher than both K-1 and K-3 ($p < 0.05$ for both TS and VS). When comparing K group and H group with the same composite microbial agents, the degradation rates of VS were improved by 12.0 %, 13.0 % and 17.1 % for the compound microbial agents H-1, H-2 and H-3, respectively. This enhancement in organic matter (VS) removal was consistent with the trends in Total Solids (TS) reduction. Furthermore, this study investigated the effects of different types of MNBs water (Air or N_2) on the biological pretreatment with bacteria or fungi. For the bacterial pretreatment with S1, Air-MNBs (A-1) yielded significantly higher TS and VS degradation rates compared to N_2-MNBs (N-1) ($p < 0.01$, Fig. 2a, b). Furthermore, For the fungal pretreatment with S2 and S3, a significant difference of VS degradation rate was similarly found between Air-MNBs (12.0 %) and N_2-MNBs (7.4 %). These results demonstrate the prominent augmentation effect of Air-MNBs on straw biodegradation(He et al., 2022).

Variations of sCOD and ORP in different treatment groups were monitored (Fig. c–f). With the increase in treatment time, the dynamic changes of dissolved oxygen and the release of reaction fermentation products may both affect the ORP (Wang et al., 2020). The significantly lower ORP in fungal-inoculated reactors (A-2 and N-2) is consistent with the high oxygen consumption rates typically associated with dense fungal mycelial networks. This creates a microaerobic to anaerobic microenvironment that can selectively favor certain lignocellulolytic enzymes, particularly those produced by white-rot fungi like Phanerodontia chrysosporium that maintain activity under oxygen-limited conditions (Etemadi-Khah and Pourbabaee, 2025; Lambooij et al., 2017; Xiong et al., 2022). The stable, moderated ORP profiles observed in MNBs-augmented systems (Fig. 2e–f) suggest that MNBs help maintain a balanced redox environment, potentially preventing the complete oxygen depletion that could stall aerobic degradation processes. It is also noteworthy that while the maximum VS reduction rate was observed in A-1, the highest sCOD level was found in N-1. This result is also corresponding to the previous statement, the stimulated activity of cellulose-decomposing bacteria also leads to the consumption of some soluble glucose (Sutaoney et al., 2024). In fact, special attention should be paid to the recovery rate of hydrolysis products, as it is usually directly proportional to the methane production in the subsequent digestion process.

Although a direct control with conventional macrobubbles was not included, the significant enhancements observed are likely attributable to the unique properties of MNBs. Unlike macrobubbles that rapidly escape the liquid, MNBs provide sustained gas transfer and can generate reactive oxygen species upon collapse (Xiao et al., 2025; Chuenchart et al., 2021). The stable, moderated ORP profiles (Fig. e-f) and the substantial increases in methane yield support the role of these MNBs-specific mechanisms, which cannot be achieved by short-lived macrobubbles.

3.3. Methane production potential test of straw after biological pretreatment

The influence of the combination of MNBs and mixed microbial agents on AD was further evaluated by the methane production potential test, which lasted from 7 days to 14 days. Fig. 3a presents the methane yield from straw pretreated with composite microbial consortia. The H-2 consortium demonstrated the highest methane production at 35.5 mL/g VS, representing an 8-fold increase over the control, followed by H-3 at 15.1 mL/g VS. The H-1 consortium yielded the lowest methane production, which is a direct consequence of its inferior pretreatment performance, as shown by its lowest VS degradation rate (Fig. 1b) and cellulose degradation efficiency (Fig. 1c). The daily progression of methane production for all pretreatment groups, which further illustrates the differential startup and accumulation patterns, is provided in Fig. S2 (Supplementary Information). The strong correlation between solids degradation during pretreatment and ultimate methane yield underscores that the primary role of the microbial consortia is to create bioavailable substrates for the anaerobic community. These production trends validate the conclusion from Section 3.1 that a balanced synergy (as achieved in H-2), rather than a simply high fungal proportion, is crucial for optimizing the entire process from straw decomposition to biomethane production. This pattern aligns with the "fungal-dominated decomposition and bacterial-mediated exploitation" model (Miao et al., 2025), wherein the fungi primarily break down the complex lignocellulose, and the bacteria, in a complementary role, facilitate the conversion of intermediates to ultimately promote methanogenesis. The balanced ratio in H-2/K-2 optimally harnesses this synergy, leading to superior system performance (Thongbunrod and Chaiprasert, 2022).

Fig. 3b and c compared the cumulative methane production from straw pretreated with Air-MNBs and N_2-MNBs. As shown, the K-2 group (S1:S2:S3 = 2:1:1) achieved the highest methane production at 47.5 mL/

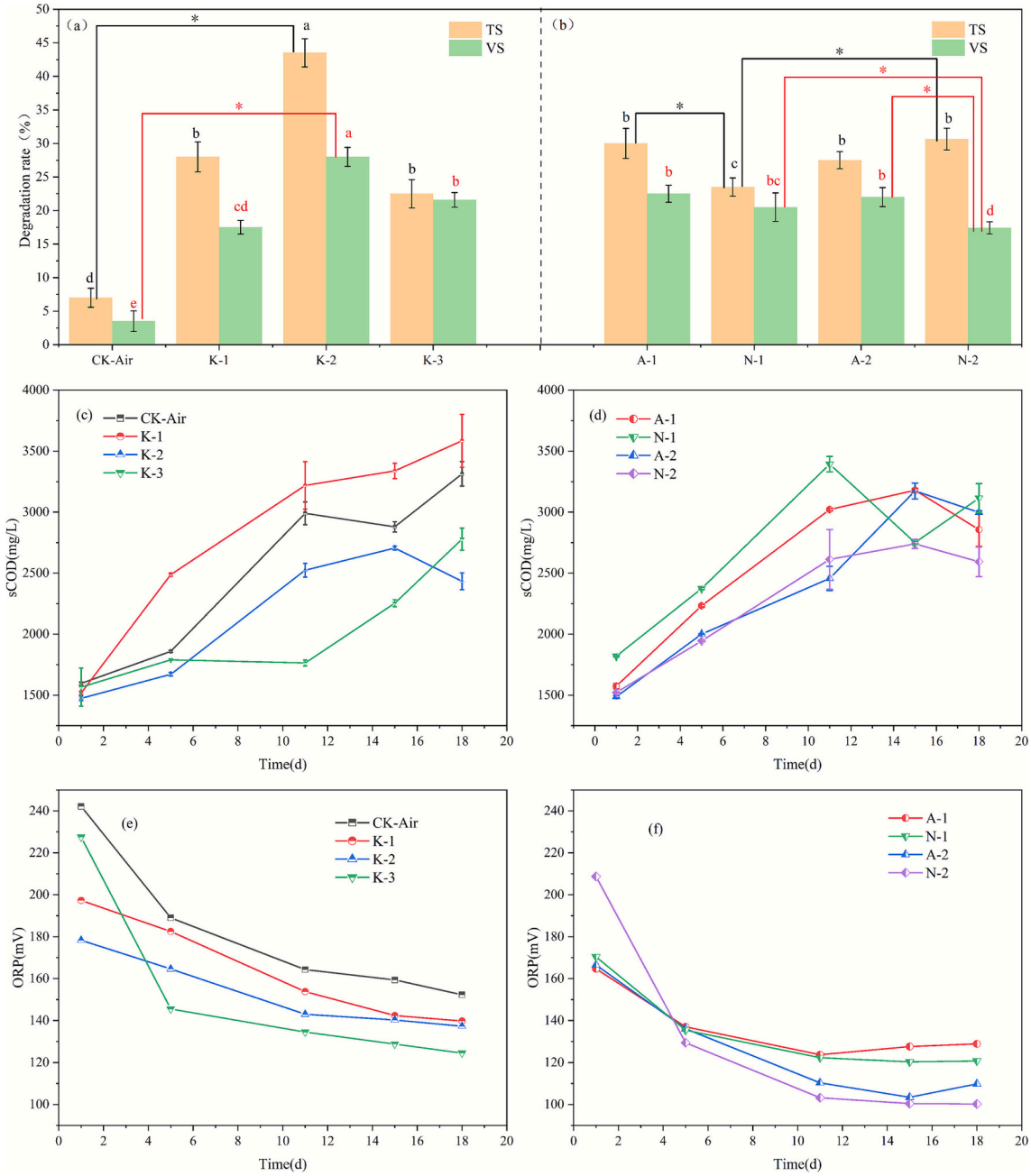


Fig. 2. Straw hydrolysis enhanced by micro-nano bubble (MNBs) augmentation. (a, c, e) Air-MNBs pretreatment with compound microbial consortia: K-1 (S1:S2:S3 = 1:1:1), K-2 (S1:S2:S3 = 2:1:1), and K-3 (S1:S2:S3 = 1:2.5:2.5). (b, d, f) Effects of different MNBs types: A-1 (S1 with Air-MNBs), N-1 (S1 with N_2-MNBs), A-2 (S2:S3 = 1:1 with Air-MNBs), and N-2 (S2:S3 = 1:1 with N_2-MNBs). (a, b) Degradation rates of TS (%) and VS (%); (c, d) soluble chemical oxygen demand, sCOD (mg/L); (e, f) oxidation-reduction potential, ORP (mV). Data represent mean ± standard deviation ($n = 3$). For each subplot (a-f), bars marked with different lowercase letters are significantly different from each other according to Tukey's HSD post-hoc test ($p < 0.05$).

g VS, which was significantly greater than that of K-3 and K-1 ($p < 0.01$). Notably, the introduction of Air-MNBs enhanced methane yields by 34.2 %, 34.0 % and 47.7 %, respectively, with the same ratio of mixed microbial agents. Meanwhile, as seen in Fig. 3c, for the bacterial pretreatment (S1), the methane yield with Air-MNBs (A-1, 31.1 mL/g VS) was significantly higher than with N_2-MNBs (N-1, 26.5 mL/g VS) ($p < 0.05$). Similarly, for the fungal pretreatment (S2:S3), Air-MNBs (A-2) also led to a significantly greater methane yield compared to N_2-MNBs (N-2) ($p < 0.05$).

Generally, these improvements correlate well with the VS degradation rates during the pretreatment phase. The incorporation of Air-MNBs has been demonstrated to significantly enhance methane yield in AD processes (Li et al., 2014; Wang et al., 2018). This phenomenon is mechanistically explained by two distinctive properties of air-MNBs, i.e. their superior mass transfer efficiency and the potential to generate reactive oxygen species (ROS). It is hypothesized that the synergistic effects of ROS-induced mild oxidative disruption and enhanced mass transfer kinetics facilitate the hydrolysis of macromolecular organic substrates (particularly polysaccharides and proteins), consequently increasing VFAs production (Yamaguchi et al., 2021; Zhang et al., 2017; Zhou et al., 2025). Similar results are observed in this study, evidenced by the higher VFAs level in A-1 and A-2 as compared to N-1 and N2. The comprehensive time-course profiles of polysaccharides, proteins, and VFAs that underpin this conclusion are detailed in Fig. S3

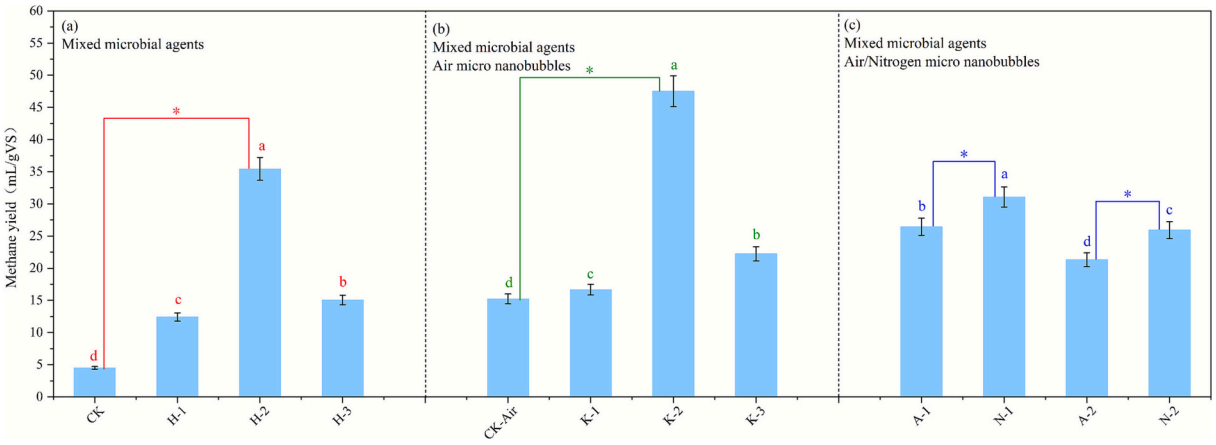


Fig. 3. Methane production potential from pretreated rice straw. (a) Composite microbial consortium pretreatments: H-1 (S1:S2:S3 = 1:1:1), H-2 (S1:S2:S3 = 2:1:1), H-3 (S1:S2:S3 = 1:2.5:2.5). (b) Air-MNBs augmented pretreatments: K-1, K-2, K-3 (microbial ratios corresponding to H-groups). (c) Comparison of MNBs types: A-1 (S1 with Air-MNBs), N-1 (S1 with N_2-MNBs), A-2 (S2:S3 = 1:1 with Air-MNBs), N-2 (S2:S3 = 1:1 with N_2-MNBs). All values represent cumulative methane yield (mL/g VS) over the digestion period. Data shown as mean ± standard deviation (n = 3). Within each panel (a-c), columns sharing the same lowercase letter are not statistically different (Tukey's HSD test, $p < 0.05$).

(Supplementary Information). Furthermore, the dissolved oxygen fraction from air establishes microaerobic conditions that stimulate metabolic activity in facultative hydrolytic/acidogenic bacterial communities, thereby indirectly augmenting methanogenic performance. These findings are consistent with previous studies reported in the literature (Lyu et al., 2024; Yan et al., 2023).

Furthermore, although the effect of MNBs augmentation on the VS degradation rate was not prominent in the fungal pretreatment (as shown in Fig. 2b), its application was still crucial as it significantly enhanced methane production in the subsequent AD stage (Fig. 3c). This indicates that MNBs improve the biodegradability and metabolic accessibility of the pretreated straw beyond what is reflected by the VS degradation metric alone. Some studies have utilized MNBs containing N_2, CO_2, and O_2 as soaking agents and AD accelerators in combination

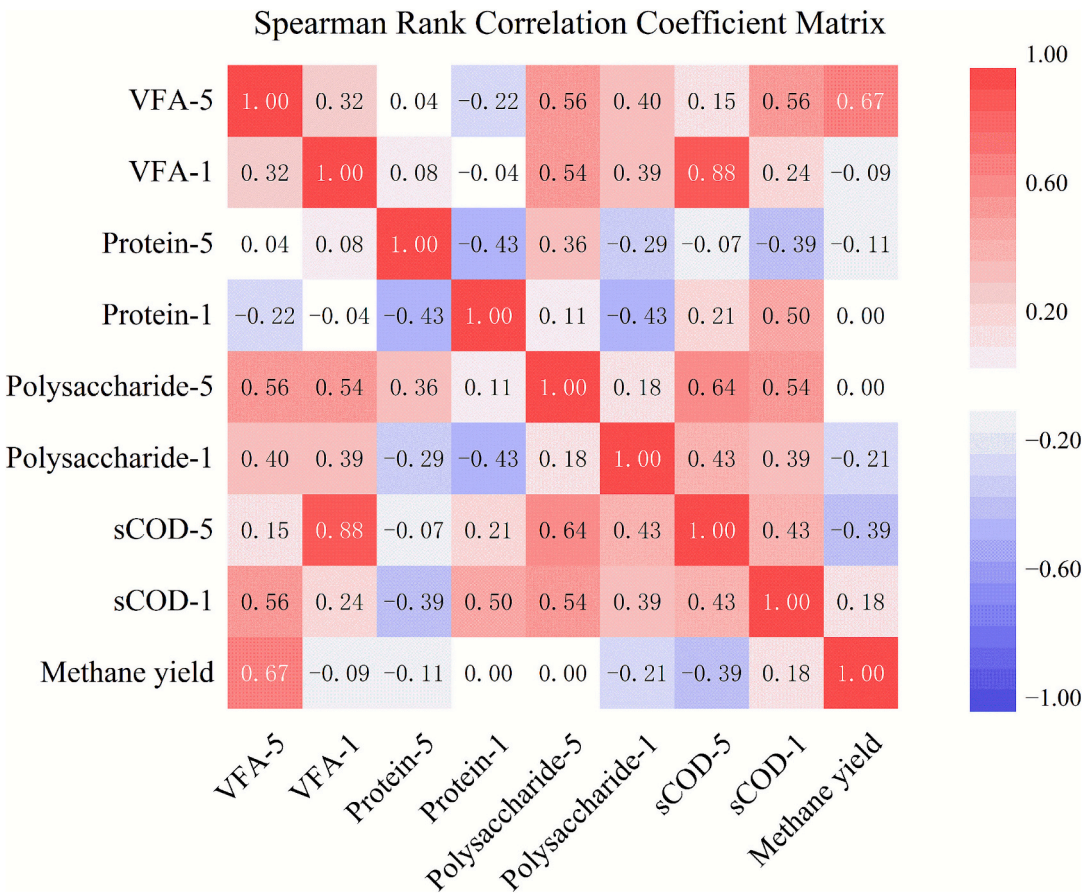


Fig. 4. Spearman correlation analysis between methane yield and soluble intermediate products during anaerobic digestion. VFA-1, protein-1, polysaccharide-1, and sCOD-1 represent concentrations measured on day 1; VFA-5, protein-5, polysaccharide-5, and sCOD-5 represent concentrations measured on day 5. Color intensity indicates correlation strength.

with activated sludge for the treatment of rice straw (Wu et al., 2019). During the two-step AD process, and the methane production from the treated rice straw increased by 11.0 %–21.4 % compared to that of the untreated straw (Kim et al., 2018; Wang et al., 2023).

To understand the relationship between hydrolysis products and methane yield, the correlation analysis was performed between methane production rate and the concentration changes of soluble intermediate products (sCOD, polysaccharides, proteins, and VFAs) on day 1 and day 5 of AD, the full time-series data for which are visualized in Fig. S3. As shown in Fig. 4, VFAs-5 exhibited a strong positive correlation with methane yield, indicating that VFAs accumulation by day 5 served as a critical indicator of successful pretreatment and hydrolysis, providing abundant substrates for subsequent methanogenesis, which aligns with previous studies (Duan et al., 2025; Wang et al., 2021). In contrast, sCOD-5 demonstrated a moderate negative correlation with methane production. This inverse relationship suggests a potential imbalance in the anaerobic microbial community during the critical acidogenic-methanogenic transition. A high sCOD level at this later stage (day 5) likely reflects an accumulation of hydrolysis and acidogenesis products, including volatile fatty acids (VFAs). This accumulation could inhibit methanogenic archaea due to acidification (pH drop) or specific VFA toxicity, ultimately leading to a lag in methane production and lower

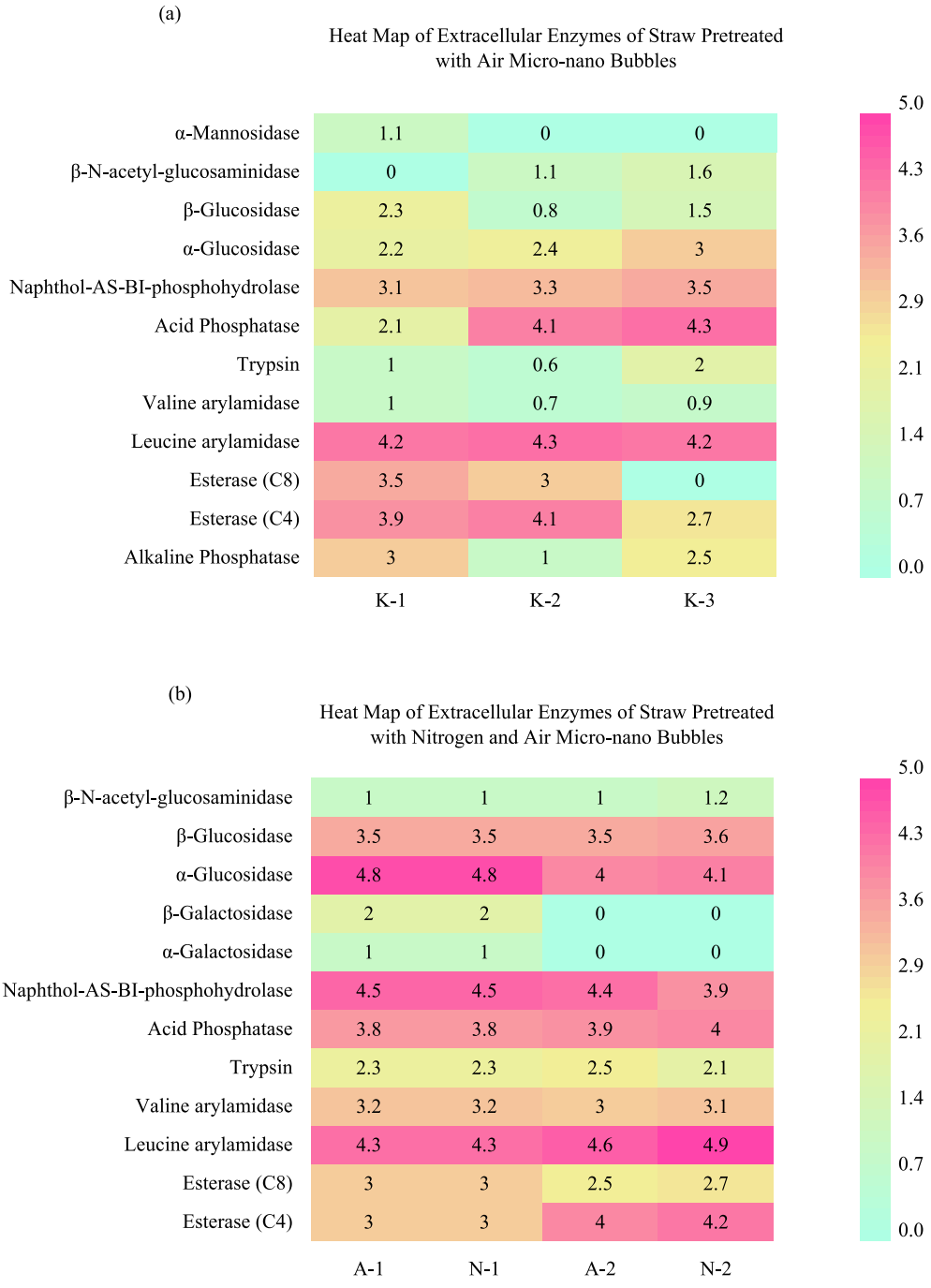


Fig. 5. Heat maps of extracellular enzyme activities. (a) Straw pretreated with Air-MNBs and compound microbial consortia: K-1 (S1:S2:S3 = 1:1:1), K-2 (S1:S2:S3 = 2:1:1), K-3 (S1:S2:S3 = 1:2.5:2.5). (b) Straw pretreated with different MNBs types: A-1 (S1 with Air-MNBs), N-1 (S1 with N_2-MNBs), A-2 (S2:S3 = 1:1 with Air-MNBs), N-2 (S2:S3 = 1:1 with N_2-MNBs). Enzyme activities are semi-quantitatively expressed on a scale from 0 (negative) to 5 (high activity). Color gradient represents relative enzyme activity levels.

final yield.

A pivotal finding emerged when we specifically interrogated the role of MNBs. The strength of the positive correlation between VFAs-5 and methane yield was significantly more pronounced in the Air-MNBs augmented groups compared to the non-MNBs controls. This provides direct statistical evidence that Air-MNBs enhance the metabolic conversion efficiency of hydrolysis products into methane, a core mechanism of the observed synergy. The marked strengthening of correlations for VFAs and polysaccharides from day 1 to day 5 further demonstrates that MNBs promote methane production by optimizing the intermediate metabolism throughout the digestion process (Hou et al., 2025; Zhang et al., 2022).

3.4. Analysis of enzyme activity of straw pretreated by composite microbial agents

The analysis of the enzyme activity of straw pretreated with a combination of MNBs and mixed microbial agents was conducted to examine the differences in enzyme activity among different types of MNBs. As shown in Fig. 5a, 11 extracellular enzymes were detected in both K-1 and K-2, while only 10 were found in K-3. All three groups shared eight enzymes: Alkaline phosphatase, Esterase(C4), Leucine arylamidase, trypsin, acid phosphatase, naphthol-AS-BI-phosphate hydrolase, α-glucosidase, and β-glucosidase—a rate-limiting enzyme in cellulose degradation that converts cellobiose to glucose, directly providing substrates for subsequent acidogenesis and methanogenesis. Notably, the detection of β-glucosidase, along with other cellulolytic (e.g., β-galactosidase) and ligninolytic enzymes, creates a complementary enzymatic portfolio that efficiently deconstructs the lignocellulosic matrix into fermentable sugars, which are precursors for VFAs and ultimately methane(Li et al., 2021). The enhanced diversity and activity of these key hydrolytic enzymes in MNBs-assisted groups, particularly β-glucosidase, correlates with the observed improvements in methane yield, suggesting MNBs create conditions favorable for the synthesis and secretion of enzymes critical for the AD feedstock preparation. No significant difference was observed between K-1 and K-2 ($p > 0.05$), but significant differences were detected between K-2 and K-3 ($p < 0.01$) and between K-1 and K-3 ($p < 0.01$) (Wang and Li, 2016). The superior straw degradation efficiency and higher sCOD levels observed in the K-1 and K-2 groups (Fig. 2a, c) can be attributed to this more diverse and balanced suite of hydrolytic enzymes, which facilitates a more comprehensive breakdown of the straw's polymeric structure into fermentable sugars (Li et al., 2024).

Fig. 5b shows the extracellular enzyme heatmap of straw pretreated with air and N_2-MNBs. It can be seen from the figure that the number of detectable enzyme activities in Groups A-1, A-2, N-1, and N-2 are 12, 10, 12, and 10 respectively. α-Galactosidase and β-galactosidase were only detected in the system with the addition of strain S1. The detection of these hemicellulose-degrading (galactosidases) and cellulose-degrading (glucosidases) enzymes aligns with the functional roles of the inoculated strains and supports the component-specific degradation patterns shown in Fig. 1c. By comparison, the number of detectable enzyme activities of strain H1 (*Bacillus*) was significantly more than that of strains S2 and S3 (*Phanerodontia*) ($p < 0.01$). Microbial cultivation analysis revealed that strains S1, S2, and S3 produced 3, 8, and 9 distinct extracellular enzymes, respectively. The analysis of extracellular enzymes provides crucial insights into the mechanistic role of MNBs.

While control groups without microbial inoculation showed negligible enzyme activity, the profiling of experimental groups revealed that MNBs augmentation led to a significant enhancement in both the diversity and activity of extracellular enzymes compared to what is typically observed in standard microbial cultivations. The sustained and diverse enzyme activities detected throughout the pretreatment phase (Fig. 5) indicate that the inoculated microbial consortia not only remained viable but were functionally active, effectively colonizing the straw substrate. This enhancement is evident in two key aspects: First, the number of detectable enzyme types was higher in the optimal consortia (e.g., 11 types in K-1 and K-2) compared to others (9 in K-3). Second, the overall enzyme activity levels were elevated in MNBs-assisted groups. Thus, the improvement can be attributed to the unique microenvironment induced by MNBs, such as elevated dissolved oxygen levels and the generation of mild oxidative stress, which can activate microbial stress responses and regulate gene expression, thereby promoting the synthesis and secretion of extracellular enzymes (Wang et al., 2024). Although the chemical properties of air and nitrogen differ, the similar enzyme response suggests that the upregulation is primarily driven by the shared physical effects of MNBs, including gas-liquid interface dynamics and associated shear forces.

The efficiency of the MNBs-microbial pretreatment approach can be contextualized by comparing it with conventional methods reported in the literature. For instance, chemical pretreatments like alkaline or acid (e.g., NaOH, H_2SO_4) treatments of rice straw typically achieve lignin degradation rates of 30–60 % but often generate inhibitory byproducts and require extensive washing, increasing operational costs and environmental impact (Anacleto et al., 2024; Tulun and Bilgin, 2019). Physical methods such as steam explosion can achieve high cellulose accessibility but are energy-intensive. In comparison, the present MNBs-microbial pretreatment achieved a 43.5 % TS degradation and a 47.5 mL/g VS methane yield, which is competitive with or superior to many conventional methods. For example, a study on fungal pretreatment of rice straw with *Phanerochaete chrysosporium* alone reported a methane yield of ~35 mL/g VS (Shah et al., 2025), while the MNBs augmentation here pushed the yield significantly higher. The principal advantage of the MNBs approach lies in its operational simplicity, minimal chemical input, and synergistic enhancement of biological activity, offering a more sustainable alternative.

The findings of this study extend beyond a technical demonstration, offering insights into synergistic mechanisms and prompting consideration of real-world application. The superior performance of MNBs-augmented pretreatment is attributed to a combination of physical and chemical effects. While enhanced oxygen mass transfer is a key factor, the unique properties of MNBs, such as their surface charge and the generation of ROS during collapse, likely play crucial roles. These properties can improve microbial-substrate contact efficiency and induce mild oxidative disruption of the lignocellulosic matrix, thereby facilitating enzymatic attack. The altered extracellular enzyme profiles observed in this study support the view of MNBs as biochemical modulators. Translating this technology to an industrial scale requires addressing its practical feasibility. The primary consideration is the energy cost of MNBs generation, which must be weighed against the significant gains in methane yield and potential reductions in pretreatment time. A major advantage lies in its straightforward integration as a pre-treatment "booster" for existing anaerobic digesters, while minimizing infrastructural changes. Future research should prioritize pilot-scale trials to validate long-term stability and conduct detailed techno-economic analyses, comparing this green approach against conventional, often more energy-intensive or chemically intensive, pretreatment methods.

4. Conclusions

In conclusion, this study establishes that augmenting designed microbial consortia with Air-MNBs presents a highly effective and novel strategy for pretreating rice straw, leading to dramatically enhanced anaerobic digestion performance. The consortium with a balanced ratio (K-2, S1:S2:S3 = 2:1:1) achieved the most pronounced synergy, resulting in the highest degradation of total solids (43.5 %) and volatile solids (28.0 %), and a maximum methane yield of 47.5 mL/g VS. The pretreatment efficacy was highly dependent on the microbial composition; Air-MNBs provided a more substantial boost to the bacterial pretreatment, whereas their effect on the fungal consortium was less distinct but still crucial for improving downstream methane production.

Furthermore, MNBs augmentation enriched the repertoire of extracellular enzymes, facilitating a more comprehensive breakdown of the straw substrate. This study provides proof-of-concept at the laboratory scale. Future work should validate these findings in pilot-scale systems and include a techno-economic analysis to assess the commercial viability of this integrated and promising pretreatment technology.

CRediT authorship contribution statement

Suyun Xu: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Yakun Lu:** Writing – original draft, Methodology, Investigation, Data curation. **Xuezhi Liu:** Methodology, Investigation. **Ruixiang Zou:** Validation, Investigation. **Yuchen Zhang:** Writing – review & editing, Software. **Mukesh Kumar Awasthi:** Writing – review & editing, Software. **Hewen Wei:** Writing – review & editing, Resources. **Hongbo Liu:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biteb.2025.102484.

Data availability

Data will be made available on request.

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